PAU 2014

Criteris específics de correcció i qualificació per ser fets públics un cop finalitzades les proves **Tecnologia industrial**

SÈRIE 5

Primera part

Exercici 1

Q1 a **Q2** c **Q3** b **Q4** d **Q5** d

Exercici 2

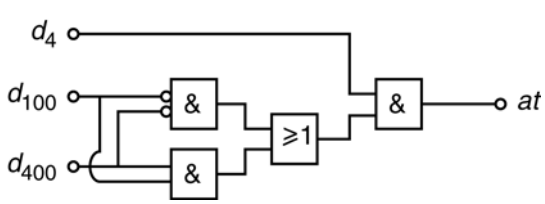
a)

d_4	d_{100}	d_{400}	at
0	0	0	0
0	0	1	X ← No és possible
0	1	0	X ← No és possible
0	1	1	X ← No és possible
1	0	0	1
1	0	1	X ← No és possible
1	1	0	0
1	1	1	1

b) Amb X = 0

$$at = d_4 \cdot \bar{d}_{100} \cdot \bar{d}_{400} + d_4 \cdot d_{100} \cdot d_{400}$$
$$= d_4 \left(\bar{d}_{100} \cdot \bar{d}_{400} + d_{100} \cdot d_{400} \right)$$

c)



Segona part

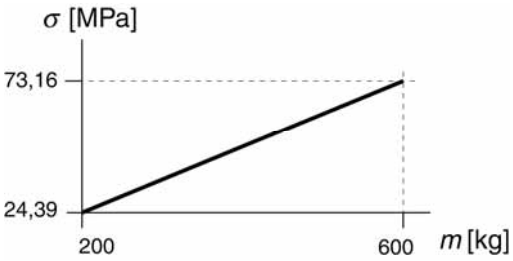
OPCIÓ A

Exercici 3

a) $L_2 = 2L_1 \cos\alpha \;\rightarrow\; \alpha = \arccos\dfrac{L_2}{2L_1} = \arccos\dfrac{6}{2 \cdot 5} = 53{,}13^{o}$

b) $\sum F = 0 \;\rightarrow\; 2F\sin\alpha - mg = 0 \;\rightarrow\; F = \dfrac{g}{2 \cdot \sin\alpha} m = 6{,}129\, m$ N, m en kg

c) $\sigma = \dfrac{F}{s} = \dfrac{F}{\pi\left(\dfrac{d}{2}\right)^2} = 121{,}9 \cdot 10^3 m$ Pa, m en kg



d) $\varepsilon = \dfrac{\sigma}{E} = \dfrac{121{,}9 \cdot 10^3 \cdot 450}{50 \cdot 10^9} = 1{,}097 \cdot 10^{-3}$

Exercici 4

a) $m_a = V \cdot \rho = 530 \cdot 10^3 \cdot 1 = 530 \cdot 10^3$ kg

$E_{dia} = m_a \, c_e \, \Delta T = 530 \cdot 10^3 \; 4{,}187 \cdot 10^3 (28 - 20) = 17{,}75 \text{ GJ} = 4931 \text{ kW h}$

b) $m_b = \dfrac{E_{dia}}{\eta \, p_b} = \dfrac{17{,}75 \cdot 10^9}{0{,}78 \cdot 12{,}54 \cdot 10^6} = 1815 \text{ kg}$

$E_{dis} = m_b \cdot p_b \, (1 - \eta) = 1815 \cdot 12{,}54 \cdot 10^6 (1 - 0{,}78) = 5{,}007 \text{ GJ} = 1391 \text{ kW h}$

c) $q = \dfrac{V}{t} = \dfrac{530 \cdot 10^3}{16 \cdot 3600} = 9{,}201 \text{L/s}$

OPCIÓ B

Exercici 3

a) $R_{mín} = \left(\dfrac{1}{R_3} + \dfrac{1}{R_4} \right)^{-1} = \dfrac{R_3 \cdot R_4}{R_3 + R_4} = 787{,}5 \; \Omega$

b) $I = \dfrac{U}{R_{mín}} = 292{,}1 \text{mA}$

c) $P_1 = \dfrac{U^2}{R_{mín}} = \dfrac{230^2}{787{,}5} = 67{,}17 \text{ W}$; $P_2 = \dfrac{U^2}{R_3} = \dfrac{230^2}{1400} = 37{,}79 \text{ W}$

$P_3 = \dfrac{U^2}{R_4} = \dfrac{230^2}{1800} = 29{,}3\hat{8} \text{ W}$; $P_4 = \dfrac{U^2}{R_3 + R_4} = \dfrac{230^2}{3200} = 16{,}53 \text{ W}$

d) $E_{elèctr} = P_1 \cdot t = 67{,}17 \cdot 2 = 134{,}3 \text{ W h}$

Exercici 4

a) $\Gamma_s = \dfrac{P_s}{\omega} = \dfrac{150 \cdot 10^3}{3800 \dfrac{2\pi}{60}} = 376{,}9 \text{ Nm}$

b) $c = \dfrac{V \cdot \rho}{P_s \cdot t} = \dfrac{600 \cdot 0{,}85}{150 \cdot 19{,}5} = 0{,}1744 \dfrac{\text{kg}}{\text{kW h}}$

c) $\eta = \dfrac{P_s}{P_e} = \dfrac{P_s}{\dfrac{p_c}{\rho} P_s \, c} = \dfrac{\rho}{p_c \, c} = \dfrac{0{,}85}{41700 \cdot 0{,}1744 \dfrac{1}{3600}} = 0{,}4209$